convolution takes place and the final image will be distorted. If you are having trouble understanding these concepts, go back and review Chapter 18, where the one-dimensional case is discussed in more detail.

The FFT algorithm is used to transform (a) and (b) into the frequency domain. This results in *four* 128×128 arrays, the real and imaginary parts of the two images being convolved. Multiplying the real and imaginary parts of (a) with the real and imaginary parts of (b), generates the real and imaginary parts of (c). (If you need to be reminded how this is done, see Eq. 9-1). Taking the Inverse FFT completes the algorithm by producing the final convolved image.

The value of each pixel in a correlation image is a measure of how well the target image matches the searched image *at that point*. In this particular example, the correlation image in (c) is composed of noise plus a single bright peak, indicating a good match to the target signal. Simply locating the brightest pixel in this image would specify the detected coordinates of the face. If we had not used the edge detection modification on the target signal, the peak would still be present, but much less distinct.

While correlation is a powerful tool in image processing, it suffers from a significant limitation: the target image must be exactly the same *size* and *rotational orientation* as the corresponding area in the searched image. Noise and other variations in the amplitude of each pixel are relatively unimportant, but an exact spatial match is critical. For example, this makes the method almost useless for finding enemy tanks in military reconnaissance photos, tumors in medical images, and handguns in airport baggage scans. One approach is to correlate the image *multiple times* with a variety of shapes and rotations of the target image. This works in principle, but the execution time will make you loose interest in a hurry.

A Closer Look at Image Convolution

Let's use this last example to explore two-dimensional convolution in more detail. Just as with one dimensional signals, image convolution can be viewed from either the *input side* or the *output side*. As you recall from Chapter 6, the input viewpoint is the best description of how convolution works, while the output viewpoint is how most of the mathematics and algorithms are written. You need to become comfortable with both these ways of looking at the operation.

Figure 24-14 shows the input side description of image convolution. Every pixel in the input image results in a scaled and shifted PSF being added to the output image. The output image is then calculated as the sum of all the contributing PSFs. This illustration show the contribution to the output image from the point at location [r,c] in the input image. The PSF is shifted such that pixel [0,0] in the PSF aligns with pixel [r,c] in the output image. If the PSF is defined with only positive indexes, such as in this example, the shifted PSF will be entirely to the lower-right of [r,c]. Don't